

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
---	---

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date: Krishna Mishra

07/07/26

(Signature and Date in Handwritten)



Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session	2025-26 (ODD)	Course Name	LAB:Neural Networks and Deep Learning
Semester	7 th	Course Code	23ADS1704
Roll No	41	Name of Student	Krishna Mishra

Practical Number	01
Course Outcome	CO1: Understand the fundamental concepts of deep learning and neural network architectures CO2: Apply different techniques to develop and optimize deep learning models. CO3: Analyze the performance of deep learning models using appropriate evaluation techniques. CO4: Design and implement deep learning solutions for real-world problems using modern tools and frameworks.
Aim	To understand the Google Colab environment and implement basic vector addition and matrix multiplication using Python and NumPy.
Theory	What is Google Colab? Google Colaboratory (Google Colab) is a free cloud-based platform developed by Google that allows users to write and execute Python programs directly in a web browser. Unlike traditional programming environments, Google Colab does not require software installation. Everything runs on Google's cloud servers. It is widely used for: ☐ Artificial Intelligence (AI)



Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

☐ Machine Learning

☐ Deep Learning

Advantages of Google Colab

☐ Free to use

☐ No installation required

☐ Works in browser

☐ GPU and TPU support

☐ Easy sharing

☐ Direct integration with Google Drive

☐ GitHub support

☐ Pre-installed Python libraries

Components of Google Colab

When students open Google Colab, they will see:

1. Notebook Name

2. Menu Bar

3. Toolbar

4. Code Cell

5. Text Cell

6. Runtime

7. Files Section

8. Variables

9. RAM and Disk Usage

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

Procedure and Execution**Algorithm**

1. Import NumPy library and create vectors and matrices in Google Colab.
2. Perform vector addition using NumPy array operations and display the result.
3. Create two matrices and perform matrix multiplication using np.matmul() or @ operator.
4. Display the multiplication result and verify the output.
5. Understand basic vector and matrix operations used in Machine Learning and Deep Learning.

Code

```
import numpy as np
vector1=np.array([1,3,5,6])
vector2=np.array([4,6,8,7])
print("vector1",vector1)
print("vector2",vector2)
result=vector1 + vector2
print(result)
```

```
import numpy as np
matrix1=np.array([[2,5,3],[4,0,2]])
matrix2=np.array([[5,2],[1,4],[3,7]])
print("matrix1",matrix1)
print("matrix2",matrix2)
result=np.matmul(matrix1,matrix2)
print(result)
```

Output 1:

```
vector1 [1 3 5 6]
vector2 [4 6 8 7]
[ 5  9 13 13]
```

Output 2:

```
matrix1 [[2 5 3]
[4 0 2]]
matrix2 [[5 2]
```

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

	<pre>[1 4] [3 7]] [[24 45] [26 22]]</pre>
Output Analysis	1. The program successfully executed in Google Colab using the NumPy library. 2. Vector addition was performed correctly by adding corresponding elements of two vectors. 3. Matrix multiplication was successfully carried out using NumPy operations. 4. The obtained outputs verify the correctness of basic vector and matrix operations. 5. These operations are fundamental concepts used in Machine Learning and Deep Learning.
Link of student Github profile where lab assignment has been uploaded	krish-na45/Neural-Network-Deep-Learning-Practicals
Conclusion	The experiment successfully implemented vector addition and matrix multiplication using Python and NumPy in Google Colab . It helped in understanding basic mathematical operations on arrays and matrices, which are important concepts used in Machine Learning and Deep Learning algorithms . The Google Colab environment provided an easy platform to write, execute, and analyze Python programs without requiring local software installation.



Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Plag Report
(Similarity index < 12%)



Page 2 of 5 - Integrity Overview

Submission ID: 27005:130402451

11% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

▸ Bibliography

Exclusions

▸ 6 Excluded Matches

Match Groups

- 2 Not Cited or Quoted 11%**
Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations 0%**
Matches that are still very similar to source material
- 0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 11% Internet sources
- 11% Publications
- 11% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Date

9/07/2026